四. 不定积分

1 原函数与不定积分

定义 设$f(x)$在区间I上有定义，若$F'(x)=f(x)$，则称$F(x)$为$f(x)$的一个原函数.

定理 若$F(x)$为$f(x)$在区间I上的一个原函数，则$f(x)$在I上的任一原函数均可表式为$F(x)+C$. C为常数.

定义 设$F(x)$为$f(x)$在区间I上的一个原函数，则$f(x)$的全体原函数为$F(x)+C$称为函数$f(x)$的不定积分，记为
$$\int f(x)dx = F(x)+C.$$

定理 若$f(x)$在区间I上连续，则$f(x)$有原函数.

定理 若$f(x)$在区间I上有原函数，则$f(x)$在I上有介值性且无第一类间断点.

(1) $\int 0\mathrm{d}x = C$;

(2) $\int 1\mathrm{d}x = x + C$;

(3) $\int x^\mu \mathrm{d}x = \frac{1}{\mu+1}x^{\mu+1} + C \quad (\mu \neq -1)$;

(4) $\int \frac{1}{x}\mathrm{d}x = \ln|x| + C$;

(5) $\int a^x \mathrm{d}x = \frac{a^x}{\ln a} + C \quad (a>0, a\neq 1)$;

(6) $\int e^x \mathrm{d}x = e^x + C$;

(7) $\int \sin x\mathrm{d}x = -\cos x + C$;

(8) $\int \cos x\mathrm{d}x = \sin x + C$;

(9) $\int \sec^2 x\mathrm{d}x = \int \frac{1}{\cos^2 x}\mathrm{d}x = \tan x + C$;

(10) $\int \csc^2 x dx = \int \frac{1}{\sin^2 x} dx = -\cot x + C$;

(11) $\int \sec x \tan x dx = \sec x + C$;

(12) $\int \csc x \cot x dx = -\csc x + C$;

(13) $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$;

(14) $\int \frac{dx}{1+x^2} = \arctan x + C$.

(15) $\int \tan x dx = -\ln|\cos x| + C$;

(16) $\int \cot x dx = \ln|\sin x| + C$;

(17) $\int \sec x dx = \ln|\sec x + \tan x| + C$;

(18) $\int \csc x dx = \ln|\csc x - \cot x| + C$;

(19) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$;

(20) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$;

(21) $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$;

(22) $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln\left|x + \sqrt{x^2 \pm a^2}\right| + C$;

(23) $\int \sqrt{a^2-x^2} dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$;

(24) $\int \sqrt{x^2 \pm a^2} dx = \frac{x}{2}\sqrt{x^2 \pm a^2} \pm \frac{a^2}{2}\ln\left|x + \sqrt{x^2 \pm a^2}\right| + C$.

以上各式中的 $a>0$.

不定积分的性质

$\left(\int f(x)dx\right)' = f(x)$, $\int [af(x) + bf(x)]dx = a\int f(x)dx + b\int f(x)dx$

2 换元积分法

换元积分公式

设 $f(x)$ 连续, $x=\varphi(t)$ 有连续导数.

$$\int f(\varphi(t))\varphi'(t)dt = \int f(\varphi(t))d\varphi(t) \xlongequal{x=\varphi(t)} \int f(x)dx \quad (*)$$

第一换元积分法

若 $\int f(\varphi(t))\varphi'(t)dt$ 不易计算，通过变换 $x=\varphi(t)$，由(*)式化为 $\int f(x)dx$ 来计算，积分后将 $x=\varphi(t)$ 代入.

例. (1) $\int \frac{\arctan x}{1+x^2}dx = \int \arctan x \, d\arctan x = \frac{1}{2}\arctan^2 x + C$

(2) $\int \tan x\,dx = \int \frac{\sin x}{\cos x}dx = \int -\frac{1}{\cos x}d\cos x = -\ln|\cos x| + C$

(3) $\int \frac{1}{\sqrt{a^2-x^2}}dx = \int \frac{\frac{1}{a}}{\sqrt{1-(\frac{x}{a})^2}}dx = \int \frac{1}{\sqrt{1-(\frac{x}{a})^2}}d\frac{x}{a} = \arcsin\frac{x}{a} + C$

(4) $\int \sin^2 x\cos^3 x\,dx = \int \sin^2 x\cos^2 x\,d\sin x = \int (\sin^2 x - \sin^4 x)d\sin x$

$= \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C$

(5) $\int \frac{x}{x^2+1}dx = \int \frac{1}{2}\cdot\frac{1}{x^2+1}dx^2 = \frac{1}{2}\arctan x^2 + C$

(6) $\int \frac{1}{x^2-a^2}dx = \int \frac{(x+a)-(x-a)}{2a(x+a)(x-a)}dx = \frac{1}{2a}\int (\frac{1}{x-a} - \frac{1}{x+a})dx$

$= \frac{1}{2a}(\ln|x-a| - \ln|x+a|) + C = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$

(7) $\int \sec x\,dx = \int \frac{1}{\cos x}dx = \int \frac{\cos x}{\cos^2 x}dx = -\int \frac{1}{\sin^2 x-1}d\sin x$

$= -\frac{1}{2}\ln\left|\frac{\sin x-1}{\sin x+1}\right| + C = \ln|\sec x+\tan x| + C$

(8) $\int \csc x\,dx = \int \frac{1}{\sin x}dx = \int \frac{\sin x}{\sin^2 x}dx = -\int \frac{1}{\sin^2 x}d\cos x$

$= \int \frac{1}{\cos^2 x-1}d\cos x = \frac{1}{2}\ln\left|\frac{\cos x-1}{\cos x+1}\right| + C = \ln|\csc x-\tan x| + C$

(9) $\int \tan^3 x\sec^3 x\,dx = \int \tan^2 x\sec^2 x\,d\sec x = \int (\sec^2 x-1)\sec^2 x\,d\sec x$

$$= \int (\sec^4 x - \sec^2 x)\,d\sec x = \frac{1}{5}\sec^5 x - \frac{1}{3}\sec^3 x + C$$

第二换元积分法

若$\int f(x)dx$不易计算，通过变换 $x=\varphi(t)$，由(*)式化为$\int f(\varphi(t))\varphi'(t)dt$

计算，积分后将 $t=\varphi^{-1}(x)$代入.

例: (1) $\int (\arcsin x)^2 dx \xlongequal{t\leftrightarrow\arcsin x} \int t^2 d\sin t = t^2\sin t - \int 2t\sin t\,dt$

$= t^2\sin t + 2t\cos t - 2\int \cos t\,dt$

$= t^2\sin t + 2t\cos t - 2\sin t + C$

$= x\arcsin^2 x + 2\sqrt{1-x^2}\arcsin x - 2x + C$

(2) $\int \frac{\sqrt{x}-1}{\sqrt[3]{x}+1}dx \xlongequal{\sqrt[6]{x}\leftrightarrow t} \int \frac{t^3-1}{t^2+1}dt^6 = 6\int \frac{t^8-t^5}{t^2+1}$

$= 6\int (t^6-t^4-t^3+t^2+t-1+\frac{1-t}{t^2+1})dt$

$= 6(\frac{t^7}{7}-\frac{t^5}{5}-\frac{t^4}{4}+\frac{t^3}{3}+\frac{t^2}{2}-t)-6(\frac{1}{2}\int\frac{1}{t^2+1}d(t^2+1)+\int\frac{1}{t^2+1}dt)$

$= 6(\frac{t^7}{7}-\frac{t^5}{5}-\frac{t^4}{4}+\frac{t^3}{3}+\frac{t^2}{2}-t) - 3\ln(t^2+1) + 6\arctan t + C$

$= 6(\frac{x^{\frac{7}{6}}}{7}-\frac{x^{\frac{5}{6}}}{5}-\frac{x^{\frac{2}{3}}}{4}+\frac{x^{\frac{1}{2}}}{3}+\frac{x^{\frac{1}{3}}}{2}-x^{\frac{1}{6}}) - 3\ln(x^{\frac{1}{3}}+1)$

$+ 6\arctan x^{\frac{1}{6}} + C$

(3) $\int \frac{1}{\sqrt{x^2+a^2}}dx \xlongequal{x\leftrightarrow a\tan t} \int \frac{1}{\sqrt{a^2(\tan^2 t+1)}}da\tan t = \int \sec t\,dt$

$= \ln|\sec t+\tan t| + C = \ln|x+\sqrt{x^2+a^2}| + C$

(4) $\int\sqrt{x^2-a^2}\,dx \xlongequal{x\leftrightarrow a\sec t} \int a\tan t\, d\,a\sec t = a^2\int \sec t\tan^2 t\, dt$

$= a^2\int \sec^3 t\, dt - a^2\int \sec t\, dt$

$= \frac{1}{2}a^2\tan t\sec t - \frac{1}{2}a^2\ln|\sec t+\tan t| + C$

$= \frac{1}{2}x\sqrt{x^2-a^2} - \frac{1}{2}a^2\ln|x+\sqrt{x^2-a^2}| + C$

(5) $\int\sqrt{a^2-x^2}\,dx \xlongequal{x\leftrightarrow a\sin t} \int a\cos t\, d\,a\sin t = a^2\int \cos^2 t\, dt$

$= a^2\int \frac{1+\cos 2t}{2}\,dt = \frac{1}{2}a^2 t + \frac{1}{4}a^2\sin 2t + C$

$= \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{1}{2}x\sqrt{a^2-x^2} + C$

(6) $\int\sqrt{5-4x-4x^2}\,dx = \int\sqrt{6-(2x+1)^2}\,dx \xlongequal{2x+1\leftrightarrow\sqrt{6}\sin t} \int \sqrt{6}\cos t\, d\sqrt{6}\sin t$

$= 6\int \cos^2 t\, dt = 3t + \frac{3}{2}\sin 2t + C$

$= 3\arcsin\frac{2x+1}{\sqrt{6}} + \frac{2x+1}{2}\sqrt{6-(2x+1)^2}\,dx + C$

(7) $\int\frac{\sqrt{a^2-x^2}}{x^4}\,dx \xlongequal{x\leftrightarrow\frac{1}{t}} -\int t\sqrt{a^2t^2-1}\,dt = -\frac{1}{2a^2}\int\sqrt{a^2t^2-1}\,d\,a^2t^2$

$= -\frac{1}{2a^2}\cdot\frac{2}{3}(a^2t^2-1)^{\frac{3}{2}} + C$

$= -\frac{1}{3a^2}(a^2t^2-1)^{\frac{3}{2}} + C$

注. 含有$\sqrt{a^2-x^2}$的. 作变换 $x=a\sin t$

含有$\sqrt{a^2+x^2}$的. 作变换 $x=a\tan t$

含有$\sqrt{x^2-a^2}$的. 作变换 $x=a\sec t$

含有$\sqrt{ax+b}$的. 作变换 $\sqrt{ax+b}=t$.

3 分部积分

分部积分公式

$$\int f(x)g'(x)dx=\int f(x)\,dg(x)=f(x)g(x)-\int f'(x)g(x)\,dx$$

$$=f(x)g(x)-\int g(x)\,df(x)$$

例：(1) $\int (x^2+x)e^x dx = (x^2+x)e^x-\int(2x+1)e^x dx$

$$=(x^2+x)e^x-\left[(2x+1)e^x-\int 2e^x dx\right]$$

$$=(x^2-x-1)e^x+2e^x+C$$

$$=(x^2-x+1)e^x+C$$

(2) $\int x^2\sin x\,dx=-x^2\cos x+\int 2x\cos x\,dx$

$$=-x^2\cos x+2x\sin x-\int 2\sin x\,dx$$

$$=2x\sin x+(2-x^2)\cos x+C$$

(3) $\int \ln x\,dx = x\ln x-\int \frac{1}{x}\cdot x\,dx$

$$=x\ln x-x+C$$

(4) $\int x^2\ln^2 x\,dx=\frac{1}{3}x^3\ln^2x-\int\frac{1}{3}x^3\frac{2\ln x}{x}dx$

$$=\frac{1}{3}x^3\ln^2x-\frac{2}{3}\int x^2\ln x\,dx$$

$$=\frac{1}{3}x^3\ln^2x-\frac{2}{3}\left[\frac{1}{3}x^3\ln x-\int\frac{1}{3}x^3\cdot\frac{1}{x}dx\right]$$

$$=\frac{1}{9}x^3\ln^2x-\frac{2}{9}x^3\ln x+\frac{2}{9}\int x^2dx$$

$$=\frac{1}{9}x^3\ln^2x-\frac{2}{9}x^3\ln x+\frac{2}{27}x^3+C$$

(5) $\int \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} - \int -\frac{1}{x} \cdot \frac{1}{x} dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx$

$= -\frac{\ln x}{x} - \frac{1}{x} + C$

(6) $\int e^{2x} \sin x \, dx = -e^{2x} \cos x + 2\int e^{2x} \cos x \, dx$

$= -e^{2x} \cos x + 2[e^{2x} \sin x - 2\int e^{2x} \sin x \, dx]$

即 $\int e^{2x} \sin x \, dx = \frac{1}{5} e^{2x} (2\sin x - \cos x) + C$

(7) $\int \sec^3 x \, dx = \int \sec^2 x \cdot \sec x \, dx = \tan x \sec x - \int \sec x \cdot \tan x \cdot \tan x \, dx$

$= \tan x \sec x - \int \sec x (\sec^2 - 1) dx$

$= \tan x \sec x - \int \sec^3 x \, dx + \int \sec x \, dx$

即 $\int \sec^3 x \, dx = \frac{1}{2}(\tan x \sec x + \int \sec x \, dx)$

$= \frac{1}{2} \tan x \sec x + \frac{1}{2} \ln|\sec x + \tan x| + C$

(8) $I_n = \int \sec^n x \, dx$, $n \in N^*$

$I_1 = \int \sec x \, dx = \ln|\sec x + \tan x| + C$

$I_2 = \int \sec^2 x \, dx = \tan x + C$

$n \geq 3$, $\int \sec^n x \, dx = \int \sec^{n-2} x \sec^2 x \, dx$

$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-3} x \sec x \tan^2 x \, dx$

$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) dx$

$= \sec^{n-2} x \tan x - (n-2) \int \sec^n x \, dx + (n-2) \int \sec^{n-2} x \, dx$

$I_n = \frac{\sec^{n-2} x \tan x}{n-1} - \frac{n-2}{n-1} I_{n-2}$.

(9) $I_n = \int \sin^n x\,dx$

$I_0 = x + C$, $I_1 = -\cos x + C$.

$n \geq 2$, $I_n = \int \sin^{n-1}x \sin x\,dx = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cos^2 x\,dx$

$$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\,(1-\sin^2 x)\,dx$$

$$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\,dx - (n-1)\int \sin^n x\,dx$$

$$I_n = -\frac{\sin^{n-1}x\cos x}{n} + \frac{n-1}{n} I_{n-2}$$

(10) $\int \arctan x\,dx = x\arctan x - \int \frac{x}{1+x^2}\,dx$

$$= x\arctan x - \int \frac{1}{2}\cdot\frac{1}{1+x^2}\,d(1+x^2)$$

$$= x\arctan x - \frac{1}{2}\ln(1+x^2) + C$$

(11) $\int \arcsin x\,dx = x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\,dx$

$$= x\arcsin x + \int \frac{1}{2\sqrt{1-x^2}}\,d(1-x^2)$$

$$= x\arcsin x + \sqrt{1-x^2}$$

(12) $\int \arcsin^2 x\,dx = x\arcsin^2 x - \int \frac{2x}{\sqrt{1-x^2}}\arcsin x\,dx$

$$= x\arcsin^2 x - 2\sqrt{1-x^2}\arcsin x + \int 2\sqrt{1-x^2}\cdot\frac{1}{\sqrt{1-x^2}}\,dx$$

$$= x\arcsin^2 x - 2\sqrt{1-x^2}\arcsin x + 2x + C$$

(13) $\int \sqrt{x^2-a^2}\,dx = x\sqrt{x^2-a^2} - \int \frac{x^2}{\sqrt{x^2-a^2}}\,dx$

$$= x\sqrt{x^2-a^2} - \int \frac{x^2-a^2+a^2}{\sqrt{x^2-a^2}}\,dx$$

$$= x\sqrt{x^2-a^2} - \int \sqrt{x^2-a^2}\,dx - \int \frac{a^2}{\sqrt{x^2-a^2}}\,dx$$

$$= x\sqrt{x^2-a^2} - \int \sqrt{x^2-a^2}\,dx - a^2\ln|x+\sqrt{x^2-a^2}| + C$$

即 $$\int \sqrt{x^2-a^2}\,dx = \frac{1}{2}x\sqrt{x^2-a^2} - \frac{1}{2}a^2\ln|x-\sqrt{x^2-a^2}| + C$$

注: $$\int \sec^n x\,dx = \int \sec^{n-2}x\,d\tan x$$

$$\int \csc^n x\,dx = \int \csc^{n-2}x\,d(-\cot x)$$

$$\int \sin^n x\,dx = \int \sin^{n-1}x\,d(-\cos x)$$

$$\int \cos^n x\,dx = \int \cos^{n-1}x\,d\sin x$$

4 特殊函数的积分

有理函数 $R(x) = \frac{P(x)}{Q(x)}$，即由两个多项式的商表示的函数.

定理 设 $R(x)=\frac{P(x)}{Q(x)}$ 为最简真分式，且 $Q(x)$ 在实数域上的质因式分解式为 $Q(x) = b_0(x-a)^\lambda \cdots (x^2+px+q)^\mu \cdots$ ，$\lambda, \mu \in N^*$,

则 $R(x)$ 可唯一分解为

$$R(x) = \frac{A_1}{(x-a)^\lambda} + \frac{A_2}{(x-a)^{\lambda-1}} + \cdots + \frac{A_\lambda}{x-a} + \cdots + \frac{M_1x+N_1}{(x^2+px+q)^\lambda}$$
$$+ \frac{M_2x+N_2}{(x^2+px+q)^{\lambda-1}} + \cdots + \frac{M_\lambda x+N_\lambda}{x^2+px+q} + \cdots$$

例: $$\int \frac{x-3}{(x-1)(x^2-1)}dx = \int\left(\frac{1}{x-1} - \frac{1}{(x-1)^2} - \frac{1}{x+1}\right)dx$$
$$= \ln|x-1| + \frac{1}{x-1} - \ln|x+1| + C$$
$$= \ln\left|\frac{x-1}{x+1}\right| + \frac{1}{x-1} + C$$

$$\int \frac{x}{(x^2+1)^2(x^2+x+1)}dx = \int \frac{x^2+x+1-(x^2+1)}{(x^2+1)^2(x^2+x+1)}dx$$

$$=\int\frac{1}{(x^2+1)^2}dx-\int\frac{1}{(x^2+1)(x^2+x+1)}dx$$

$$=\frac{x}{(x^2+1)^2}+\int\frac{2x}{(x^2+1)^3}dx-\int\frac{x}{x^2+1}dx+\int\frac{x+1}{x^2+x+1}dx$$

$$=\frac{x}{(x^2+1)^2}+\int\frac{1}{(x^2+1)^3}dx^2-\frac{1}{2}\int\frac{1}{x^2+1}d(x^2+1)+\int\frac{x+1}{x^2+x+1}dx$$

$$=\frac{x}{(x^2+1)^2}-\frac{1}{2}\frac{1}{(x^2+1)^2}-\frac{1}{2}\ln|x^2+1|+\int\frac{x+1}{x^2+x+1}dx$$

其中 $\int\frac{x+1}{x^2+x+1}dx=\frac{1}{2}\int\frac{1}{x^2+x+1}d(x^2+x+1)-\frac{1}{2}\int\frac{1}{x^2+x+1}dx$

$$=\frac{1}{2}\ln|x^2+x+1|-\frac{1}{2}\int\frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}dx$$

$$=\frac{1}{2}\ln|x^2+x+1|-\frac{1}{2}\cdot\frac{4}{3}\cdot\frac{\sqrt{3}}{2}\int\frac{1}{\frac{4}{3}(x+\frac{1}{2})^2+1}d\,\frac{2}{\sqrt{3}}(x+\frac{1}{2})$$

$$=\frac{1}{2}\ln|x^2+x+1|-\frac{\sqrt{3}}{3}\arctan\frac{2\sqrt{3}}{3}(x+\frac{1}{2})+C$$

从而 $\int\frac{x}{(x^2+1)^2(x^2+x+1)}dx=\frac{x}{(x^2+1)^2}-\frac{1}{2(x^2+1)^2}+\frac{1}{2}\ln\left|\frac{x^2+x+1}{x^2+1}\right|-\frac{\sqrt{3}}{3}\arctan\frac{2\sqrt{3}}{3}(x+\frac{1}{2})+C$

三角函数有理式

通过半角代换化为有理函数

令 $t=\tan\frac{x}{2}$，则 $\sin x=\frac{2t}{1+t^2}$， $\cos x=\frac{1-t^2}{1+t^2}$

例： $\int\frac{1+\sin x}{(1+\cos x)\sin x}dx \xlongequal{t=\tan\frac{x}{2}} \int\frac{t^2+2t+1}{2t}dt$

$$=\frac{1}{2}\int(t+\frac{1}{t}+2)dt=\frac{1}{4}t^2+\frac{1}{2}\ln|t|+t+C$$

$$=\frac{1}{4}\tan^2\frac{x}{2}+\frac{1}{2}\ln|\tan\frac{x}{2}|+\tan\frac{x}{2}+C$$

例：(1) 设 $f(x)=\begin{cases} \ln x, & x\geq 1 \\ \frac{1}{2}-\frac{1}{1+x^2}, & x<1 \end{cases}$. 求 $\int f(x)dx$.

$\int \ln x\,dx = x\ln x - x + C_1$. $\int (\frac{1}{2}-\frac{1}{1+x^2})dx = \frac{1}{2}x - \arctan x + C_2$

$-1+C_1 = \frac{1}{2}-\frac{\pi}{4}+C_2$, $C_1 = \frac{3}{2}-\frac{\pi}{4}+C_2$

则 $\int f(x)dx = \begin{cases} x\ln x - x + \frac{3}{2}-\frac{\pi}{4}+C, & x\geq 1 \\ \frac{1}{2}x - \arctan x + C, & x<1 \end{cases}$

(2) 设 $x\in[0,2\pi]$,

$\int\sqrt{1+\sin x}\,dx = \int\sqrt{(\sin\frac{x}{2}+\cos\frac{x}{2})^2}\,dx = \int\left|\sin\frac{x}{2}+\cos\frac{x}{2}\right|dx$

$= \sqrt{2}\int\left|\sin(\frac{x}{2}+\frac{\pi}{4})\right|dx = \begin{cases} \sqrt{2}\int\sin(\frac{x}{2}+\frac{\pi}{4})\,dx, & x\in[0,\frac{3\pi}{2}] \\ -\sqrt{2}\int\sin(\frac{x}{2}+\frac{\pi}{4})\,dx, & x\in[\frac{3\pi}{2},2\pi] \end{cases}$

$= \begin{cases} -2\sqrt{2}\cos(\frac{x}{2}+\frac{\pi}{4})+C, & x\in[0,\frac{3\pi}{2}], \\ 2\sqrt{2}\cos(\frac{x}{2}+\frac{\pi}{4})+4\sqrt{2}+C, & x\in[\frac{3\pi}{2},2\pi]. \end{cases}$

(3) $\int\frac{2\sin x+\cos x}{\sin x-\cos x}dx$

设 $2\sin x+\cos x = a(\sin x-\cos x)+b(\sin x-\cos x)'$

$= (a+b)\sin x + (b-a)\cos x$

得 $a=\frac{1}{2}$, $b=\frac{3}{2}$.

从而 $\int\frac{2\sin x+\cos x}{\sin x-\cos x} = \int\frac{\frac{1}{2}(\sin x-\cos x)+\frac{3}{2}(\sin x-\cos x)'}{\sin x-\cos x}dx$

$= \int\frac{1}{2}dx + \int\frac{3}{2}\cdot\frac{1}{\sin x-\cos x}d(\sin x-\cos x)$

$= \frac{1}{2}x + \frac{3}{2}\ln|\sin x-\cos x| + C$

(4) $\int \frac{1}{1+\sin x} dx = \int \frac{1-\sin x}{1-\sin^2 x} dx = \int \frac{1-\sin x}{\cos^2 x} dx$

$= \int \sec^2 x \, dx + \int \frac{1}{\cos^2 x} d\cos x$

$= \tan x - \frac{1}{\cos x} + C$

(5) $\int \frac{1}{2+\sin x} dx = \int \frac{1}{2\sin^2\frac{x}{2} + 2\cos^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2}} dx$

$= \frac{1}{2}\int \frac{\frac{1}{\cos^2\frac{x}{2}}}{\tan^2\frac{x}{2} + \tan\frac{x}{2} + 1} dx = \int \frac{1}{(\tan\frac{x}{2}+\frac{1}{2})^2 + \frac{3}{4}} d(\tan\frac{x}{2}+\frac{1}{2})$

$= \frac{2\sqrt{3}}{3} \arctan \frac{2\sqrt{3}}{3}(\tan\frac{x}{2} + \frac{1}{2}) + C$

(6) $\int \frac{1}{3\sin^2 x+1} dx = \int \frac{1}{4\sin^2 x + \cos^2 x} dx = \int \frac{1}{2\tan^2 x+1} \cdot \frac{1}{\cos^2 x} dx$

$= \frac{1}{2}\int \frac{1}{2\tan^2 x+1} d2\tan x = \frac{1}{2}\arctan(2\tan x) + C$

(7) $\int \frac{1}{\sin(x+\alpha)\sin(x+\beta)} dx = \int \frac{\sin[(x+\alpha)-(x+\beta)]}{\sin(\alpha-\beta)\sin(x+\alpha)\sin(x+\beta)} dx$

$= \frac{1}{\sin(\alpha-\beta)} \int \frac{\sin(x+\alpha)\cos(x+\beta) - \cos(x+\alpha)\sin(x+\beta)}{\sin(x+\alpha)\sin(x+\beta)} dx$

$= \frac{1}{\sin(\alpha-\beta)} \int [\cot(x+\beta) - \cot(x+\alpha)] dx$

$= \frac{1}{\sin(\alpha-\beta)} \ln\left|\frac{\sin(x+\beta)}{\sin(x-\beta)}\right| + C$

(8) $\int \frac{1+x^2}{1+x^4} dx = \int \frac{\frac{1}{x^2}+1}{\frac{1}{x^2}+x^2} dx = \int \frac{1}{(x-\frac{1}{x})^2+2} d(x-\frac{1}{x})$

$= \frac{\sqrt{2}}{2}\arctan\frac{\sqrt{2}}{2}(x-\frac{1}{x}) + C$

(9) $\int \frac{1}{x(x^6+1)}dx = \int \frac{x^5}{x^6(x^6+1)}dx = \frac{1}{6}\int \frac{1}{x^6(x^6+1)}dx^6$

$= \frac{1}{6}\int (\frac{1}{x^6} - \frac{1}{x^6+1})dx^6 = \frac{1}{6}\ln\frac{x^6}{x^6+1} + C$

(10) $\int x\tan^2 x dx = \int x(\sec^2 x - 1)dx = \int x\sec^2 x dx - \int x dx$

$= \int x d\tan x - \frac{1}{2}x^2 = x\tan x - \int \tan x dx - \frac{1}{2}x^2$

$= x\tan x + \ln|\cos x| - \frac{1}{2}x^2 + C$

(11) $\int \sin x \ln\tan x dx = \int \ln\tan x d(-\cos x)$

$= -\cos x\ln\tan x + \int \csc x dx = -\cos x\ln\tan x + \ln|\csc x - \cot x| + C$

(12) $\int \frac{1+\sin x}{1+\cos x}e^x dx = \int \frac{(1+\sin x)(1-\cos x)}{1-\cos^2 x}e^x dx$

$= \int \frac{1-\cos x + \sin x - \sin x\cos x}{\sin^2 x}e^x dx$

$= \int \csc^2 x e^x dx - \int \csc x\cot x e^x dx + \int \csc x e^x dx - \int \cot x e^x dx$

$= -\int e^x d\cot x + \int e^x d\csc x + \int \csc x e^x dx - \int \cot x e^x dx$

$= -e^x\cot x + \int \cot x e^x dx + e^x\csc x - \int \csc x e^x dx + \int \csc x e^x dx - \int \cot x e^x dx$

$= (\csc x - \cot x)e^x + C$

$\int \frac{1+\sin x}{1+\cos x}e^x dx = \int \frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}e^x dx$

$= \frac{1}{2}\int \sec^2\frac{x}{2}e^x dx + \int \tan\frac{x}{2}e^x dx = \int e^x d\tan\frac{x}{2} + \int \tan\frac{x}{2}e^x dx$

$= e^x\tan\frac{x}{2} - \int e^x\tan\frac{x}{2}dx + \int e^x\tan\frac{x}{2}dx = e^x\tan\frac{x}{2} + C$

(13) $\int \frac{\arcsin x}{\sqrt{(1-x^2)^3}} dx \xlongequal{x \leftrightarrow \sin t} \int \frac{t}{\cos^3 t} d\sin t = \int t\sec^2 t\, dt$

$= \int t\, d\tan t = t\tan t - \int \tan t + C = t\tan t - \ln|\cos t| + C$

$= \frac{x}{\sqrt{1-x^2}} \arcsin x - \ln\sqrt{1-x^2} + C$

(14) $\int \sqrt{e^x+1}\, dx \xlongequal{\sqrt{e^x+1}=t} \int t\, d\ln(t^2-1) = 2\int \frac{t^2}{t^2-1} dt$

$= 2\int (1-\frac{1}{t^2-1}) dt = 2t - \ln\left|\frac{t-1}{t+1}\right| + C = 2\sqrt{e^x+1} - \ln\left|\frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1}\right| + C$

(15) $\int \sqrt{\tan x}\, dx \xlongequal{t \leftrightarrow \sqrt{\tan x}} \int t\, d\arctan t^2 = \int \frac{2t^2}{t^4+1} dt$

$= \int \frac{t^2+1}{t^4+1} dt + \int \frac{t^2-1}{t^4+1} dt = \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}} dt + \int \frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}} dt$

$= \int \frac{1}{(t-\frac{1}{t})^2+2} d(t-\frac{1}{t}) + \int \frac{1}{(t+\frac{1}{t})^2-2} d(t+\frac{1}{t})$

$= \frac{\sqrt{2}}{2} \arctan \frac{\sqrt{2}}{2}(t-\frac{1}{t}) + \frac{\sqrt{2}}{4} \ln\left|\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right| + C$

$= \frac{\sqrt{2}}{2} \arctan \frac{\sqrt{2}}{2}(\sqrt{\tan x} - \frac{1}{\sqrt{\tan x}}) + \frac{\sqrt{2}}{4} \ln\left|\frac{\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}} - \sqrt{2}}{\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}} + \sqrt{2}}\right| + C$

(16) $\int \frac{\arctan e^x}{e^{2x}} dx = -\frac{1}{2}\int \arctan e^x\, de^{-2x}$

$= -\frac{1}{2}(\frac{\arctan e^x}{e^{2x}} - \int \frac{1}{(1+e^{2x})e^x} dx) = -\frac{1}{2} \frac{\arctan e^x}{e^{2x}} + \frac{1}{2}\int \frac{de^x}{(1+e^{2x})e^{2x}}$

$= -\frac{1}{2} \frac{\arctan e^x}{e^{2x}} + \frac{1}{2}\int (\frac{1}{e^{2x}} - \frac{1}{1+e^{2x}}) de^{2x}$

$= -\frac{1}{2} \frac{\arctan e^x}{e^{2x}} + \frac{1}{2} \ln\left|\frac{e^{2x}}{1+e^{2x}}\right| + C$

(17) $\int \frac{x+1}{x(1+xe^x)} dx = \int \frac{(x+1)e^x}{xe^x(1+xe^x)} dx = \int (\frac{1}{xe^x} - \frac{1}{1+xe^x}) dxe^x$

$= \ln\left|\frac{xe^x}{1+xe^x}\right| + C$

(18) $\int \sqrt{\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}} dx = \int \sqrt{\ln(x+\sqrt{1+x^2})} d\ln(x+\sqrt{1+x^2}) = \frac{2}{3}\ln^{\frac{3}{2}}(x+\sqrt{1+x^2}) + C$

(19) $\int x^2(e^{3x} - \sqrt{4-3x^3}) dx = \int x^2 e^{3x} dx - \int x^2\sqrt{4-3x^3} dx$

$= \frac{1}{3}\int x^2 d e^{3x} - \frac{1}{9}\int \sqrt{4-3x^3} d 3x^3$

$= \frac{1}{3}x^2e^{3x} - \frac{1}{3}\int 2xe^{3x} dx + \frac{1}{9}\int \sqrt{4-3x^3} d(4-3x^3)$

$= \frac{1}{3}x^2e^{2x} - \frac{2}{3}\cdot\frac{1}{3}(xe^{3x} - \int e^{3x}dx) + \frac{2}{27}(4-3x^3)^{\frac{3}{2}}$

$= (\frac{x^2}{3} - \frac{2}{9}x + \frac{2}{27})e^{2x} + \frac{2}{27}(4-3x^3)^{\frac{3}{2}} + C$

(20) $\int \frac{(1+x^2)\arcsin x}{x^2\sqrt{1-x^2}} dx = \int \frac{\arcsin x}{\sqrt{1-x^2}} dx + \int \frac{\arcsin x}{x^2\sqrt{1-x^2}} dx$

$\xlongequal{\arcsin x = t} \int \arcsin x \, d\arcsin x + \int \frac{t}{\sin^2 t \cos t} d\sin t$

$= \frac{1}{2}\arcsin^2 x - \int t \, d\cot t = \frac{1}{2}\arcsin^2 x - t\cot t + \int \cot t \, dt$

$= \frac{1}{2}\arcsin^2 x - t\cot t + \ln|\sin t| + C$

$= \frac{1}{2}\arcsin^2 x - \sqrt{1-x^2}\arcsin x + \ln|x| + C$

(21) $\int \frac{x\ln x}{(1+x^2)^2} dx = \int \ln x \, d(-\frac{1}{2}\cdot\frac{1}{1+x^2}) = -\frac{\ln x}{2(1+x^2)} + \frac{1}{2}\int \frac{1}{x(1+x^2)} dx$

$= -\frac{\ln x}{2(1+x^2)} + \frac{1}{4}\int \frac{1}{x^2(1+x^2)} dx^2 = -\frac{\ln x}{2(1+x^2)} + \frac{1}{4}\int (\frac{1}{x^2} - \frac{1}{1+x^2}) dx^2$

$= -\frac{\ln x}{2(1+x^2)} + \frac{1}{4}\ln\frac{x^2}{1+x^2} + C$